ECA5403-ELECTROMAGNETIC FILEDS AND WAVES

S.Rupika

192512398

C01-3

EXPERIMENT 3

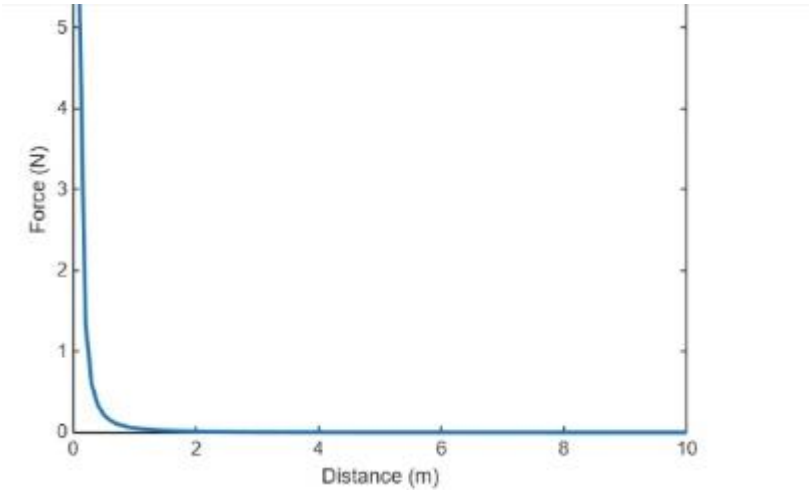
AIM:

To determine and plot the Force of attraction between two electric Charges and two parallel conductors separated by varying distance between them using MATLAB.

Test case (i) Determination and plotting of force between two electric charges with varying distance

```
 clc;
clear;
q1 = input('Enter the charge q1=');
q2 = input('Enter the charge q2=');
 epsilon = 8.854e-12; k = 1/(4*pi*epsilon);
 r = linspace(0.1,10,100); F = k*q1*q2./(r.^2);
disp('Force at r = 1 m is:'); disp(k*q1*q2/(1^2));
plot(r,F,'LineWidth',2);
 xlabel('Distance (m)');
 ylabel('Force (N)');
 title('Force between two electric charges');
```

Output:- Enter the charge q1=2e-6
Enter the charge q2=3e-6
Force at r = 1 m is: 0.0539 N
Graph: Force decreases with distance ($1/r^2$ behavior)



Test case (ii) Determination and plotting of force between two electric charges with different magnitudes

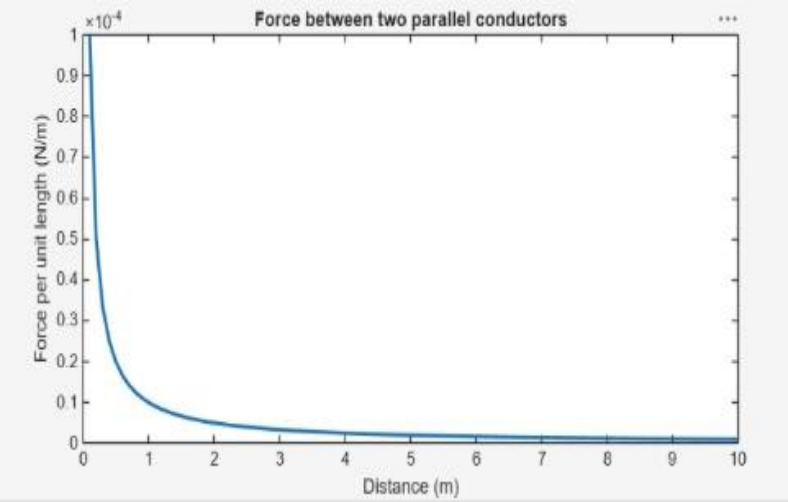
Code:

```
clc;
q1 = 5e-6;
q2 = 2e-6;
epsilon = 8.854e-12;
 k = 1/(4*pi*epsilon);
 r = linspace(0.5,5,100);
 F = k*q1*q2./(r.^2);
 plot(r,F,'LineWidth',2);
 xlabel('Distance (m)');
 ylabel('Force (N)');
title('Force between two charges (different magnitudes)');
```

Output:-

- Force magnitude increases with increase in charge
- Inverse square decay maintained

OUTPUT:



Test case (iii) Determination and plotting of force per unit length between two parallel conductors with varying distance

Code:

```
clc;

mu0 = 4*pi*1e-7;

I1 = 5;

I2 = 10;

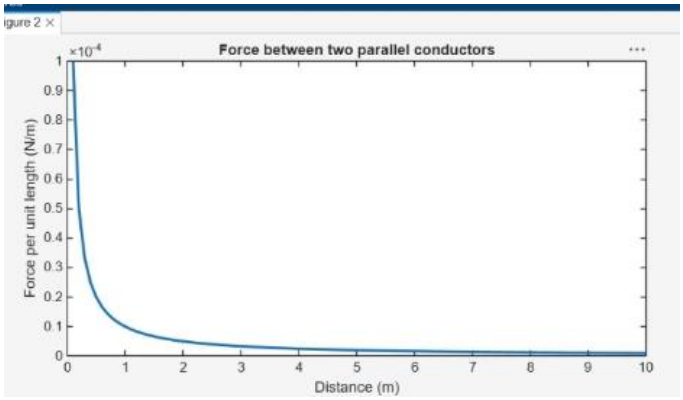
d = linspace(0.1,10,100);
```

```
F = (mu0*I1*I2)./(2*pi*d);
plot(d,F,'LineWidth',2);
xlabel('Distance (m)');
ylabel('Force per unit length (N/m)');
title('Force between two parallel conductors');
```

Output:

● Force decreases with increase in distance

OUTPUT:



Test case (iv) Determination and plotting of force between two parallel conductors carrying equal currents

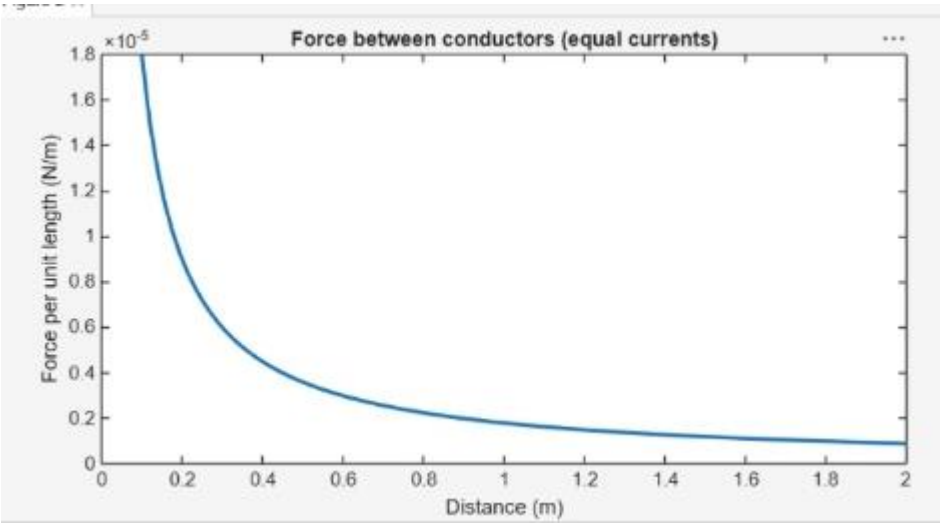
Code:

```
clc;
 mu0 = 4*pi*1e-7;
 I1 = 3;
 I2 = 3;
d = linspace(0.1,2,100);
 F = (mu0*I1*I2)./(2*pi*d);
plot(d,F,'LineWidth',2);
 xlabel('Distance (m)');
 ylabel('Force per unit length (N/m)');
title('Force between conductors (equal currents)');
```

Output:-

Force magnitude reduced compared to Test case (iii) Same inverse relationship

OUTPUT:



Objective	Assessment Criterion	Excellent – 4	Good – 3	Satisfactory – 2	Needs Improvement – 1	Weightage
Identify physical quantities and force models	Identify physical quantities and force models	Correctly identifies charge/current, separation, medium and force direction for both cases.	Identifies most quantities and models correctly.	Needs guidance in selecting the correct relation or direction.	Uses incorrect force models or quantities.	25
Calculate electric and magnetic forces	Calculate electric and magnetic forces	Implements Coulomb-force and parallel-current force calculations correctly with units.	Calculations are mostly correct with minor errors.	Some calculations are correct but require assistance.	Force calculations are invalid or absent.	25
Evaluate force versus separation	Evaluate force versus separation	Correctly varies separation, plots trends and checks theoretical distance dependence.	Trend and plots are correct with minor omissions.	Limited sweep or incomplete trend interpretation.	Does not establish a valid relationship.	30
Explain and report results	Explain and report results	Explains attraction/repulsion and reports assumptions, units and conclusions clearly.	Conclusion is correct with minor documentation gaps.	Conclusion is incomplete or partly inaccurate.	Conclusion is missing or incorrect.	20